TN266 v1p0 comments

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Inviato: martedì 3 maggio 2016 18.08
A: Davide Sgalaberna

Hi Davide,

Here's some comments on sections 1-3. Looking good - all just style/clarity points I think. I'll wait for the FC version before looking at the rest.

Cheers,
Tom

It's worth doing a global find-replace on super-K - should be Super-K

3 - Steve is now Liverpool

45 - I guess you probably will/have, but remember to change the abstract when you add the FC results. I'd also quote some headline numbers here (e.g. exclusion region of dcp @ 90%, dm32sq and theta23 best fit & 1 sigma uncertainties, hierarchy preference, …)

89 - Missing full stop before "As in"

105 - Is there a sensitivity plot showing that 2D does nothing to help numu sample @ current stats? I'd reference this here.

130 - "unique theta bin" -> "single theta bin"

Sec 2.1 - you say "lepton angle" everywhere, without distinction between "reconstructed lepton angle" and "true lepton angle". It's probably "reconstructed lepton angle" everywhere - just make sure & add it in
Also, what's beta in Eq 1?

Sec 2.2.4 - it'd be nice to have the variable cuts in the same form as 2.2.3 e.g.: (evclass == 1)

206 - Is it a fiTQun pi0 rejection cut?

303 - 14 10 degree bins from 0 to 140

306 - unique -> single

325 - don't reference table 9 (the SK error table) - it's not a BANFF output. Happens a few times

354 - Set A in Tab 12?

362 - 101-> 109 65+36+1+7(? not all 7 are systematic parameters at the same time of course, otherwise you wouldn't be measuring anything!)

377 - have a complicated -> have complicated

389 - there is no theta binning described in http://www.t2k.org/asg/oagroup/inputs/201415/2015SKsplines

Tab 12 - "which corresponds to the opposite octant." makes me think it's a reflection into the opposite octant. But it's just an arbitrary point in the opposite octant. So perhaps "which corresponds to an arbitrary point in the opposite octant."

Tab 6-9 - can you add a column with things like f^banff_17;t;r to these tables? Because they're used in Tab 13 onwards without being explicitly defined. (This was a comment in my viva, otherwise I'd of never spotted it!)
Also, in the captions say something along the lines of "Only the systematic parameters which effect this sample are shown."
Careful with overflowing onto the page numbers too!

Fig 6 - dcp is measured too, isn't it? Maybe explicitly say that this is for the reactor constraint (the envelope will be slightly smaller in the T2K-only case, since theta13 won't be thrown)
continuos -> continuous
where is the blue (prefix)?
Fig 7 - as Fig 6 and...
y-axis title should be 'fractional error' (or similar)
Fig 8-12 - as Fig 6
Fig 13-17 - as Fig 7

Can you do a clearpage (or whatever) to put Tab 17-21 before Fig 8?

Tab 17-21 - this looks overly complicated… Can you put the XSec stuff first (it's the ugliest part) and maybe describe the things in more detail (what are the Correlated XSec parameters? Are MEC (corr) and MECbar (corr) included there?) Can you also choose whether you want the 'default' assumption to be pre-fit or post-fit, state that in the table, and only use the non-assumed one e.g. (pre-fit) on every line it's relevant
I'm also rather confused - sqrt(4.15^2 + 2.92^2 + 0.03^2) is not 11.9
Style suggestion, use left alignment and some spacing to show what's a subset of what. For example (missing bits out…):
All
SKDet+FSI+SI
Flux+XSec
Flux
XSec Tot
XSec Corr Tot
MEC (corr)
NC 1gamma (uncorr)
XSec nue/numubar (is this corr or uncorr?)